1 COMBINATION

nCr = C(n,r) = number of ways to SELECT r items from n where ORDER DOES NOT MATTER (AB = BA).

nCr = n! / [r!(n−r)!] • C(6,3) = 6!/(3!3!) = 720/36 = 20

PROPERTIES: Symmetry nCr = nC(n−r) • Boundary nC0 = nCn = 1 • Pascal: nCr = (n−1)C(r−1) + (n−1)Cr • Total subsets = 2^n • nPn = n!, nP1 = n, 0! = 1.

APPLICATIONS: probability/lottery, binomial expansion coefficients, statistics, bioinformatics (genes), game theory, cryptography.

WORKED EXAMPLES

E1 choose 3 from 6: 6C3 = 20 ways.

E2 symmetry: e.g. find C(10,7): use C(10,3) = 120.

E3 Pascal: C(n,r) = C(n−1,r−1) + C(n−1,r).

E4 committee 4 from 7M+5W, exactly 2M2W: 7C2×5C2 = 21×10 = 210. (AND rule = multiply.)

E5 committee 5 from 6M+4W, at least 2 women (cases, OR rule = add): 4C2·6C3 + 4C3·6C2 + 4C4·6C1 = 120+60+6 = 186.

E6 team 4 from 10, A and B refuse together: 10C4 − (1×8C2) = 210 − 28 = 182.

E7 choose 3 letters of PROBLEM (all distinct): 7C3 = 35.

E8 committee of 3 from 4M+3W, P(≥2 women): (3C2·4C1 + 3C3)/7C3 = 13/35.

ANSWERS (exercise): select vs arrange ABC: 3P2−3C2 = 6−3 = 3 • 5M4W team of 3 ≥1 woman: 9C3−5C3 = 74 • chocolates with repetition: C(3+4−1,3) = C(6,3) = 20 • 5 from 12, 2 must be included: 10C3 = 120 • 3 balls ≥2 red from 4R5G: 4C2·5C1 + 4C3 = 24+4 = 28 • EXAM (E before A): 4!/2 = 12 • 3-digit from {1,2,3} repeat: 3^3 = 27 • MISSISSIPPI: 11!/(4!4!2!) = 34,650 • 5 in a row, 2 together: 2·4! = 48 • circular 6 beads: (6−1)!/2 = 60.

2 PERMUTATION

nPr = P(n,r): ARRANGEMENT in ORDER (ABC, ACB, BAC, BCA, CAB, CBA all different).

nPr = n!/(n−r)! • identical items (p,q,r alike): n!/(p!·q!·r!)

PROPERTIES: nPn = n! • nP1 = n • nP0 = 1 (one way to arrange nothing) • 0! = 1.

TYPES: simple / with restrictions / circular / with repetition. Circular: (n−1)! ways; if rotations equal (beads): (n−1)!/2.

APPLICATIONS: sampling without replacement, sorting/task order, DNA sequencing, scheduling/logistics, passcodes.

WORKED EXAMPLES

E1 arrange 4 of 6 students: 6P4 = 6·5·4·3 = 360.

E2 APPLE, two Ps NOT together: total 5!/2! = 60; block PP as one item → 4! = 24; answer 60−24 = 36.

E3 5 people round a circular table: (5−1)! = 24.

E4 BALLOON (7 letters, L×2, O×2): 7!/(2!·2!) = 5040/4 = 1260.

E5 3-digit numbers from {2,3,5,6,7} no repeat: 5P3 = 60.

nC2·2! = nP2 link: 10C4 = 210 committees but 10P4 = 5040 office assignments (×4! = 24).

3 PROBABILITY — CONCEPTS & PRINCIPLES

Probability measures likelihood of an event; P(E) ∈ [0,1]; 0 = impossible, 1 = certain.

P(E) = favourable outcomes / total outcomes

TERMS: Experiment = action producing outcomes. Sample space S = set of ALL outcomes. Event = subset of S. Equally likely = same probability. Mutually exclusive = cannot occur together. Exhaustive = covers all outcomes.

PROPERTIES: 0 ≤ P(A) ≤ 1 • P(S) = 1 • P(A′) = 1 − P(A) • P(A∪B) = P(A)+P(B) if mutually exclusive.

APPLICATIONS: weather, insurance/risk, games, medical tests (false +ve/−ve), ML/AI.

(A) THEORETICAL (CLASSICAL)

logic-based, no experiment. ≥1 head in 2 coins: 3/4 • die >4: 2/6 = 1/3 • red card from 52: 26/52 = 1/2.

(B) EXPERIMENTAL (EMPIRICAL)

from results. Spinner 50×: (Blue+Yellow)/50 = (10+16)/50 = 26/50 = 0.52 • coin 100 tosses, 56 heads: 56/100 = 56%.

FUNDAMENTAL RULES

Additive (mut.excl.): P(A∪B) = P(A)+P(B). General: P(A∪B) = P(A)+P(B)−P(A∩B)

Multiplication (indep.): P(A∩B) = P(A)·P(B). Complement: P(A′) = 1−P(A)

CONDITIONAL PROBABILITY

P(A|B) = P(A∩B)/P(B) ; P(B|A) = P(A∩B)/P(A)

Dependent: P(B|A) ≠ P(B). Independent: P(A∩B) = P(A)P(B).

King or Queen: 4/52+4/52 = 8/52 = 2/13.

Bag 3R2G no replacement: P(2nd green|1st red) = 2/4 = 1/2.

Coin + die, head and 4: (1/2)(1/6) = 1/12. Die not 5: 1−1/6 = 5/6.

BAYES' THEOREM

$P(A_i|B) = P(B|A_i)P(A_i) / [P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \ldots]$

Factory A,B,C = 50%,30%,20%; defects 3%,4%,5%. P(B|D) = (0.30×0.04)/(0.5×0.03+0.3×0.04+0.2×0.05) = 0.012/0.037 = 0.324 (≈32.4%).

4 RANDOM VARIABLES & DISTRIBUTION FUNCTIONS

RV = numerical outcome of a random experiment (function on sample space).

(A) DISCRETE (countable values) — PMF p(x)

p(x) ≥ 0 • Σp(x) = 1 • E(X) = μ = Σ x·p(x) • Var(X) = σ^2 = Σ(x−μ)²p = E(X^2)−μ^2 • σ = $\sqrt{\sigma^2}$

Die: E = 3.5, Var = 35/12 ≈ 2.92. 3 coins, X = heads: p = 1/8, 3/8, 3/8, 1/8; E = 1.5, Var = 0.75. PMF tables must sum to 1.

(B) CONTINUOUS — PDF f(x)

f(x) ≥ 0 • ∫f(x)dx = 1 over range • P(a≤X≤b) = ∫[a,b] f(x)dx • P(X=c) = 0 • E(X)=∫xf(x)dx • Var = ∫x^2f(x)dx − μ^2

PDF example f(x)=Cx: $\int_0^1$ Cx dx = C/2 = 1 → C = 2; E = 2/3; P(0≤X≤1/2) = 1/4.

CDF F(x) = P(X ≤ x)

discrete: F(x)=Σp(t), t≤x • continuous: F(x)=$\int_{-\infty}^{x}$ f(t)dt

PROPERTIES: 0 ≤ F ≤ 1 • F(−∞)=0, F(+∞)=1 • F is NON-DECREASING • P(X≤x) • P(a<X≤b) = F(b)−F(a) • F is right-continuous.

Die CDF: F(1)=1/6 … F(6)=1. Geometric p(x)=q^(x−1)p example: pmf at 1,2,3… with F(x)=1−q^x.

5 DISCRETE DISTRIBUTIONS

BERNOULLI (single trial, 1=success p, 0=failure q=1−p)

PMF: p(1)=p, p(0)=q • E = p • Var = pq (binomial with n=1)

Coin p=0.6 heads: P(tail)=0.4; E=0.6, Var=0.24. Exam: 9 MCQs, 1 of 6 correct: μ=np=9×(1/6)=1.5, σ^2=npq=1.25.

BINOMIAL (n fixed trials, 2 outcomes, constant p, independent)

P(X=x) = nCx·p^x·q^(n−x) • E=np • Var=npq • SD=√(npq) • Excel =BINOM.DIST(x,n,p,FALSE) / TRUE for P(X≤x)

Coin 6 tosses, exactly 2: 6C2(1/2)⁶ = 15/64 = 0.2344.

Fuse manual: n=20, p=0.05, P(X≥4) = 1−P(X≤3) = 1−BINOM.DIST(3,20,0.05,TRUE) = 0.015902.

Drug p=0.3, n=10, P(X≥9) = 0.000144 (≈1 in 7000). Table n=20, p=0.55: peak P(11)=0.1603, mean np=11, var 4.95, sd 2.2249.

POISSON (events per fixed interval; rare events, large n small p, λ=np)

P(X=x) = λ^xe^(−λ)/x! • E = Var = λ • SD = √λ • e≈2.71828 • Excel =POISSON.DIST(x,λ,FALSE

)/TRUE • P(0)=e^(−λ) • ratio p(x+1)/p(x)=λ/(x+1)
Patrol λ=1/30min: P(0)=P(1)=0.367879, P(2)=0.183940, P(≥1)=1−e⁻¹=0.632121. λ=2: P(4)=0.090224, P(≥4)=0.142877, P(<4)=0.857123. λ=3: P(5)=0.100819, P(≥5)=0.184737, P(<5)=0.815263. λ=5.5: peak P(5)=0.171, P(≤8)=0.8944. Fire 1 in 1000, 2000 houses → λ=2: P(5) = 2⁵e⁻²/5! = 0.036089. Bulbs p=0.2, n=200 → λ=40: P(<2)=e⁻⁴⁰(1+40)≈0; P(2)=800e⁻⁴⁰≈0.

GEOMETRIC (trials until FIRST success)

Type1 x=1,2,…: p(x) = q^(x−1)·p • Type2 failures before success x=0,1,…: p(x)=qˣp • E=1/p • Var=q/p² • P(X≤x)=1−qˣ • P(X≥k)=q^(k−1) • memoryless. NO Excel command — type =(1−p)^(x−1)*p
p=0.45 table: 0.45, 0.2475, 0.136125, 0.074869… (each ×0.55). p=0.65: E=1.538. p=0.54: P(4)=0.052561, P(≥4)=0.46³=0.097336, P(<4)=0.902664. p=0.60: P(5)=0.4⁴×0.6=0.01536, P(≥5)=0.0256, P(<5)=0.9744.

6 CONTINUOUS DISTRIBUTIONS

NORMAL / GAUSSIAN X~N(μ,σ²)

f(x) = (1/(σ√(2π)))·e^(−(x−μ)²/2σ²) • Z = (X−μ)/σ ~ N(0,1) • Excel =NORM.DIST(x,μ,σ,TRUE); =NORM.S.DIST(z,TRUE)
PROPERTIES: bell-shaped, symmetric about μ; mean=median=mode=μ; total area=1; 68.26% within ±1σ, 95.44% ±2σ, 99.73% ±3σ.
Z areas: P(Z≤1.5)=0.9332 • −1.96 to 0: 0.4750 • −1.96 to 1.5: 0.9082 • P(Z>2)=0.0228 • P(−2≤Z≤2)=0.9545 • P(0≤Z≤1.73)=0.4582 • P(−1<Z<1)=0.6827 • P(0≤Z≤2.5)=0.4938 • Z²<1: 0.6827; Z²<3.84146 (=±1.96): 0.9500.
Juice N(500,20): P(480−520) → z=±1 → 0.6826 = 68.26%. SAT N(1000,200), top 10%: z=1.28 → X = 1000+1.28(200) = 1256. Scores N(75,10), P(80−90): z=0.5,1.5 → 0.2417 (24.2%). N(60.57,35.15): σ=√35.15=5.9288; f(63)=0.0619, F(63)=0.6590.

EXPONENTIAL (waiting time; partner of Poisson)

f(x) = λe^(−λx), x≥0 • E = 1/λ = β • Var = 1/λ² • P(X>x) = e^(−λx) • Excel =EXPON.DIST(x,1/β,TRUE) • memoryless
e⁻² = 0.135335. Equipment 3 parts β=100: P(all≥200h OK) = P(≥2 survive) = 3p²q+p³ = 0.04999 ≈ 5%. Earthquake mean 2.4: P(X>3)=e^(−1.25)=0.286505; P(2≤X≤3)=0.148093. Poisson-exponential link: no event in 5 min ↔ wait > 5 min (both e⁻² = 0.1353).

GAMMA (waiting for k events)

f(x)= x^(α−1)e^(−x/β)/(βᵅΓ(α)) • E=αβ • Var=αβ² • mode=(α−1)β (α>1) • skew=2/√α • kurt=6/α • Γ(n)=(n−1)!, Γ(1)=1, Γ(1/2)=√π • Excel =GAMMA.DIST(x,α,β,TRUE)
Rainfall α=1.6, β=2: P(X>4) = 1−GAMMA.DIST(4,1.6,2,TRUE) = 0.289550. α=2,β=2: F(2)=0.264241, F(5)=0.712703, P(2≤X≤5)=0.448461. Mean 10, Var 25: β=Var/Mean=2.5, α=4. Fishing 1 per 0.5h, time to 4 fish ~ Gamma(4,0.5): P(2<t<4)=0.3911.

BETA (proportions on [0,1])

f(x) = x^(α−1)(1−x)^(β−1)/B(α,β) • E = α/(α+β) • Var = αβ/[(α+β)²(α+β+1)] • mode = (α−1)/(α+β−2) • median ≈ α/(α+β) approx
B(α,β) = Γ(α)Γ(β)/Γ(α+β); Γ(n)=(n−1)!. Special: α=β=1 → Uniform(0,1); α=β<1 → U-shape; α=β>1 → bell/symmetric.

UNIFORM (a,b)

f(x)=1/(b−a) • E=(a+b)/2 • Var=(b−a)²/12 • P = (length)/(b−a) • Excel =RAND()*(b−a)+a
Battery U(100,200): P(<150)=0.5, P(120−180)=0.6, P(>170)=0.3. Package U(10,30): E=20, Var=400/12=33.33, SD=5.7735. U(50,100): E=75, SD=14.434.

7 SAMPLING DISTRIBUTIONS

Sampling distribution = distribution of a STATISTIC over all possible samples. Statistic from sample; PARAMETER from population (μ, σ, p).

SAMPLE MEAN x̄ — σ KNOWN

x̄~N(μ, σ²/n): μx̄ = μ • σx̄² = σ²/n • SE = σx̄ = σ/√n • Z = (x̄−μ)/(σ/√n) ~ N(0,1)
CLT: mean of sample means = population mean (pumpkins: μ = 14 both).
Bulbs μ=800, σ=40, n=16, P(x̄<775): Z=(775−800)/(40/4)=−2.5 → P = 0.0062.
Ford 220HP, σ=15, n=4: σx̄=7.5, Z=(215−220)/7.5=−0.67 → P = 0.2514 (25.14%). n=100: σx̄=1.5, Z=−3.33 → P≈0.0004.

x̄ — σ UNKNOWN: use S (S≈σ)

Z = (x̄−μ)/(S/√n) if n>30 • t = (x̄−μ)/(S/√n) ~ t(n−1) if n≤30
Loans μ=825, x̄=780, S=105, n=35: Z=(780−825)/(105/√35) = −2.54 → P = 0.0055.

SAMPLE PROPORTION P̂

P̂~N(p, pq/n): μP = p • σP² = pq/n • Z = (P̂−p)/√(pq/n)
Finite population (no replacement): multiply by √((N−n)/(N−1)).
Variance of P̂: n=50, p=0.4 → pq/n = 0.24/50 = 0.0048. n=75, p=0.65 → 0.65×0.35/75 = 0.003.

NORMAL APPROXIMATION OF BINOMIAL (np≥5 and nq≥5)

Z = (x−np)/√(npq) • exact x: continuity correction ±0.5: P(14.5≤x≤15.5)
p=0.2, n=100, P(x≤15): Z=(15−20)/4=−1.25 → 0.1056. P(x=15): P(14.5≤x≤15.5) → Z −1.25 to −1.13 → 0.1292−0.1056 = 0.0236 (guide).
p=0.5, n=100, σP=0.05: P(0.24≤P̂≤0.42): Z −5.2 to −1.6 → 0.0548. P(P̂≥0.65): Z=3 → 1−0.9987 = 0.0013.

8 EXCEL COMMANDS & CRITICAL VALUES

=FACT(n) factorial • =PERMUT(n,r) • =COMBIN(n,r) • =BINOM.DIST(x,n,p,FALSE/TRUE) • =POISSON.DIST(x,λ,FALSE/TRUE) • =NORM.DIST(x,μ,σ,TRUE) • =NORM.S.DIST(z,TRUE) • =GAMMA.DIST(x,α,β,TRUE) • =EXPON.DIST(x,1/β,TRUE) • geometric: TYPE =(1−p)^(x−1)*p (no built-in).
Excel wants SD not variance: NORM.DIST(x,60.57,SQRT(35.15),TRUE). EXPON.DIST wants RATE 1/mean. FALSE = single value, TRUE = cumulative (≤x). At least k → 1−DIST(k−1,…,TRUE).

Z BANK (left areas Φ)

Φ(0)=0.5000 Φ(0.5)=0.6915 Φ(1)=0.8413 Φ(1.13)=0.8708 Φ(1.28)=0.8997 (top 10%) Φ(1.5)=0.9332 Φ(1.645)=0.9500 Φ(1.73)=0.9582 Φ(1.96)=0.9750 Φ(2)=0.97725 Φ(2.5)=0.9938 Φ(3)=0.9987 Φ(−z)=1−Φ(z).

WHICH DISTRIBUTION?

Count successes of n trials → BINOMIAL • count events in time/space (no limit) → POISSON • trials to 1st success → GEOMETRIC • symmetric measurement → NORMAL • waiting/lifetime → EXPONENTIAL (k events → GAMMA) • equally likely interval → UNIFORM • proportion in [0,1] → BETA.

MEAN/VARIANCE BANK

Binomial np, npq • Poisson λ, λ • Geometric 1/p, q/p² • Bernoulli p, pq • Normal μ, σ² • Exponential 1/λ, 1/λ² • Gamma αβ, αβ² • Uniform (a+b)/2, (b−a)²/12 • x̄: μ, σ²/n • P̂: p, pq/n.